

UCM & ULoG Practice Test

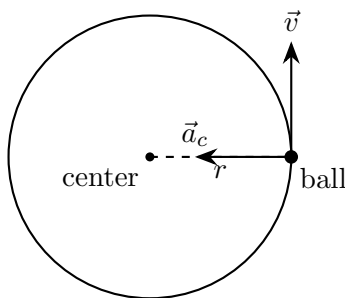
Name: _____

Date: _____

Multiple Choice: Please select the best answer choice for each of the following questions.

Questions #1–4 refer to the diagram below.

A ball moves in a horizontal circle of radius 2.0 m at a constant speed of 6.0 m/s.



- What is the magnitude of the centripetal acceleration?
 - 3.0 m/s²
 - 9.0 m/s²
 - 18 m/s²
 - 36 m/s²
- In the diagram, the direction of the centripetal acceleration $\vec{a}_c$ is:
 - Tangent to the circle, in the direction of motion.
 - Radially inward, toward the center of the circle.
 - Radially outward, away from the center of the circle.
 - Downward due to gravity.
- If the speed of the ball is increased (same radius), what happens to the centripetal acceleration?
 - It increases.
 - It decreases.
 - It remains the same.
 - It becomes zero.
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 - It remains the same.
 - It becomes zero.

$$\Sigma F_r = ma_r \quad a_c = \frac{v^2}{r} = \omega^2 r \quad v = \omega r \quad F_c = m \frac{v^2}{r} \quad F_g = G \frac{m_1 m_2}{r^2}$$

5. For an object moving in a circle at constant speed, which quantity is always changing?
 - a. Acceleration
 - b. Velocity
 - c. Mass
 - d. Radius
6. Which force is primarily responsible for keeping a satellite in orbit around Earth?
 - a. Normal force
 - b. Friction
 - c. Gravitational force
 - d. Applied force from engines at all times
7. If the distance between two masses doubles, the gravitational force between them becomes:
 - a. Twice as large
 - b. Four times as large
 - c. Half as large
 - d. One-fourth as large
8. Two objects attract each other gravitationally. If one object's mass is tripled (distance unchanged), the gravitational force:
 - a. Stays the same
 - b. Triples
 - c. Increases by a factor of 9
 - d. Decreases by a factor of 3

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Problem Solving

1. A 0.80 kg ball is swung in a horizontal circle of radius 1.5 m at a constant speed of 4.0 m/s.
 - a. What is the real force that acts as the centripetal force?
 - b. Draw a free body diagram showing all the forces acting on the object.
 - c. Compute the centripetal acceleration.
 - d. Determine the centripetal force required.
2. A 1200 kg satellite orbits a planet of mass 6.0×10^{24} kg at a distance of 7.0×10^6 m from the planet's center.
 - a. Determine the gravitational force acting on the satellite.
 - b. Determine the satellite's orbital speed for a circular orbit.
 - c. Determine the orbital period.

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3. A 60 kg rider is seated in a roller-coaster car that moves through a **vertical loop** of radius 20 m. At the **top** of the loop, the car's speed is 10 m/s.
- Draw a free body diagram for the rider at the **top** of the loop. Label F_g and F_N and indicate the direction of the centripetal acceleration.
 - Compute the centripetal acceleration at the top.
 - Determine the normal force on the rider at the **top** of the loop. State whether the rider feels lighter or heavier than normal.

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